

System Architecture Document

Comprehensive Architectural Blueprint for PricePilot AI — Frontend, Backend, ML, Database & Deployment

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Release Version:	Version 2.0.0
Platform:	PricePilot AI — AI-Powered Dynamic Pricing & Demand Forecasting SaaS
Technology Stack:	Python 3.13 · FastAPI · React 19 · Scikit-Learn · PostgreSQL · SQLAlchemy
ML Algorithm:	Extra Trees Regressor (Extremely Randomized Trees) — R^2 0.9650
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Organization:	Infosys Springboard 7.0 Internship Program
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DOCUMENT REVISION HISTORY

Rev	Date	Author	Reviewer	Description
1.0	2026-08-01	Team PricePilot	Infosys Mentor	Initial Draft — Project Inception
1.5	2026-08-04	Narendar Reddy	Manvitha	Mid-Sprint Review — ML Pipeline Integration
2.0	2026-08-07	Full Team	Academic Committee	Final Production Release — University Submission

1.0 ARCHITECTURAL OVERVIEW

PricePilot AI is architected as a decoupled, cloud-native SaaS platform. The system separates concerns across four independent tiers: Presentation (React 19 Vite SPA), Application (FastAPI ASGI microservices), Data (Neon Cloud PostgreSQL with SQLite local fallback), and Intelligence (Scikit-Learn Extra Trees Regressor serialized via joblib). This decoupled architecture enables horizontal scaling of each tier independently and facilitates CI/CD deployment pipelines.

2.0 CLOUD DEPLOYMENT ARCHITECTURE

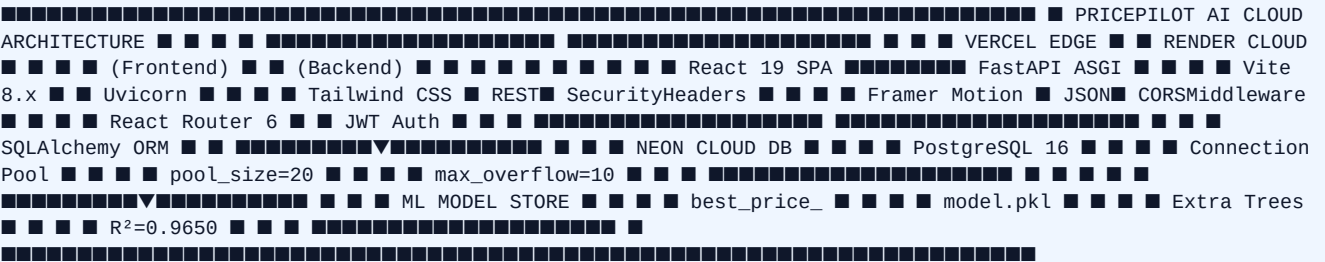


Figure 2.1 — PricePilot AI Cloud Deployment Architecture Diagram

3.0 FRONTEND ARCHITECTURE (React 19 SPA)

Layer	Technology	Version	Responsibility
UI Rendering	React	19.x	Declarative component-based UI
Build Tool	Vite	8.x	Hot module replacement, production bundling
Routing	React Router DOM	6.x	Client-side SPA navigation (17 routes)
State Management	React Context API	Built-in	AuthContext (JWT), ToastContext (notifications)
HTTP Client	Axios / Fetch	Latest	REST API calls with Bearer token injection
Animation	Framer Motion	Latest	Page transitions, glassmorphism effects
Icons	Lucide React	Latest	Consistent SVG icon system
Error Handling	ErrorBoundary	Custom	Runtime error capture and fallback UI

Table 3.1 — Frontend Technology Stack

4.0 BACKEND ARCHITECTURE (FastAPI Microservices)

Module	File	Responsibilities
Application Entry	main.py	ASGI setup, middleware, router registration, static serving
Authentication Router	routers/auth.py	Register, Login, JWT issuance, OTP generation/verification
Users Router	routers/users.py	User CRUD, bulk ops, openpyxl Excel export
Prediction Router	routers/predict.py	ML inference, prediction persistence, activity logging
Dashboard Router	routers/dashboard.py	Stats aggregation, recent activity retrieval
Docs Router	routers/docs.py	Document metadata, binary file streaming
Security Module	security.py	JWT encode/decode, Bcrypt password hashing utilities
Middleware Module	middleware.py	SecurityHeadersMiddleware injecting HSTS, X-Frame headers
Database Module	database.py	SQLAlchemy engine (pool_size=20), session factory
Schemas Module	schemas.py	Pydantic v2 request/response schema definitions
Models Module	models.py	SQLAlchemy ORM table classes (10 tables)

Table 4.1 — Backend Module Architecture

5.0 MACHINE LEARNING ARCHITECTURE

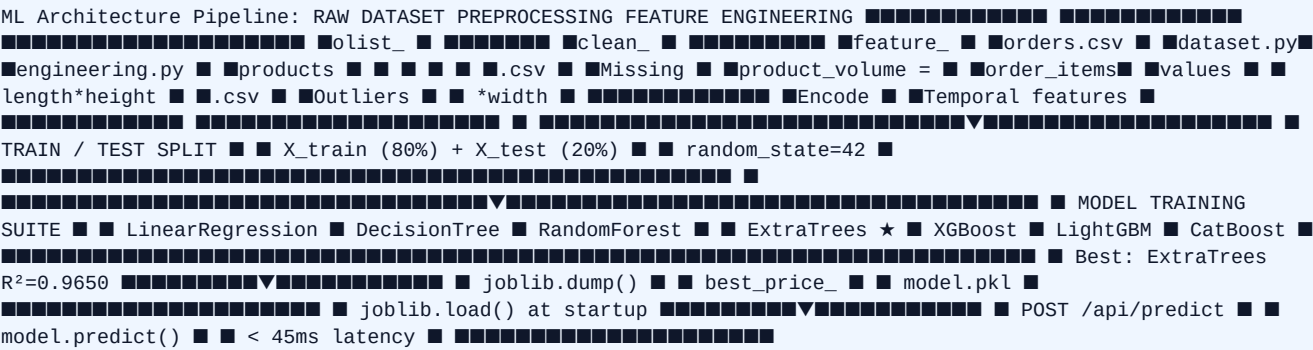


Figure 5.1 — Machine Learning Architecture Pipeline Diagram